

Number of observations

 $N = 100$ $N = 1,000$ $N = 10,000$

Time (s)

Reversible

Non-reversible

Number of states (K)

Proposed methods

Exact
Approximate

Benchmarks

Autodiff (torchtree)
Autodiff (treeflow)
PhyloGrad (only reversible)

Scaling guides

$\mathcal{O}(K^*)$
Exact ($K = 8$)

